

The 17 GHz Gap

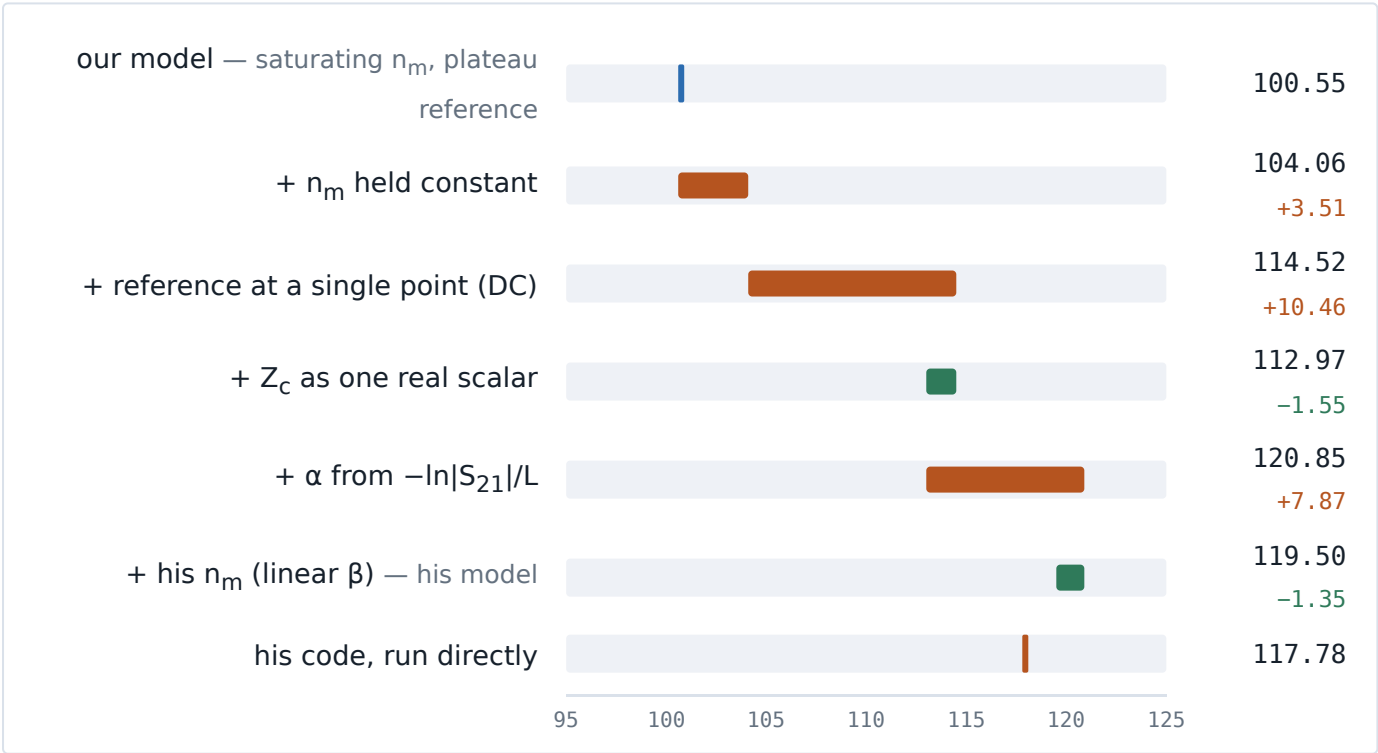
Two codes, the same Touchstone file, the same transfer function — and 100.5 GHz against 117.8 GHz. Here is every gram of that difference, weighed.

This is not a disagreement about physics. Both codes use the identical Gopalakrishnan traveling-wave expression. Feeding his extracted α , β and Z_c into our formula returns 117.33 GHz against his 117.78 — a 0.4% round-trip. The formula was never in question.

The gap is entirely in *how the line parameters are extracted from the S-parameters, and where the 0 dB reference is taken*. Four choices differ. Three of them push the answer up.

Where the 17 GHz comes from

Each row below changes exactly one thing from the row above. Nothing else moves.



EO bandwidth, GHz. The residual 1.7 GHz between the last two rows is his 0.13 GHz frequency grid, his **Circuit** round-trip, and a -3 dB search that averages two grid indices instead of interpolating.

The three that push upward

+10.5

Taking the 0 dB reference at one frequency

A mismatched electrode *rings*. The forward and backward microwave waves interfere, so the response carries a standing-wave ripple of period $c/(2 \cdot n_m \cdot L)$. For this device that is **3.96 GHz** — measured 3.96 — at **0.49 dB peak to peak**.

Anchoring the reference at one frequency anchors on one *phase* of that ripple. DC happens to sit in a trough, **-0.32 dB** below the mean level, so the whole curve lifts by 0.32 dB. Near the -3 dB point the response only falls ~0.034 dB/GHz, so that becomes ten gigahertz of apparent bandwidth.

The proof is that the answer moves when nothing physical does:

ANCHOR FREQUENCY	0	0.5	1	2	5 GHZ
bandwidth	110.3	108.1	101.9	94.3	102.2

Same file, same device, same everything — 16 GHz of spread from a setting that ought to be irrelevant. **We now average the reference over a whole number of ripple periods**, which cancels it. Spread afterwards: **0.000 GHz**.

α from $-\ln|S_{21}|/L$

This treats S_{21} as pure propagation, $e^{(-\gamma L)}$. It is not. For a line whose Z_c differs from the 50 Ω reference,

$$S_{21} = (1-\Gamma^2) \cdot e^{-\gamma L} / (1 - \Gamma^2 e^{-2\gamma L})$$

so **the mismatch loss $(1-\Gamma^2)$ is being counted as propagation loss**. With $Z_c = 60.09 \Omega$, $\Gamma = 0.0917$, and the predicted bias is a flat **+0.293 dB/cm**. Measured against the ABCD de-embedding: **+0.308**. The mechanism is confirmed, not inferred.

Now the counter-intuitive part, and the reason this *raises* the bandwidth rather than lowering it. The bias is **constant in frequency**, so when the data is fitted with $a\sqrt{f} + b \cdot f + c$, it is absorbed into **c** — and the slope **b** comes out roughly half:

FIT	A ($\sqrt{f}$)	B (F)	C	A@150	A@300
his	+0.0702	+0.0295	+0.5470	5.83	10.60
ours	0.0000	+0.0419	+0.2835	6.58	12.87

A flat offset at low frequency **flattens the extrapolation**. Over the 16.5 mm device at 150 GHz that is 9.6 dB of loss instead of 10.9 dB — and the -3 dB point moves out accordingly. He is not under-counting loss where he measured it; he is under-counting it where he extrapolated.

We use ABCD de-embedding, which inverts the full two-port and removes the mismatch exactly. On a synthetic line with known truth its error is **-0.0000 dB/cm**, against **+0.038** for the $|S_{21}|$ method — which matches that line's own predicted mismatch bias of +0.030.

Holding n_m constant

Fitting $\text{Im}(\gamma)$ linearly forces n_m to a constant. That is an *assumption*, not a measurement — and it is the mildest of the three. It is also, on the evidence, a good assumption for this device.

The way to settle it: fit each model on the lower half of the measured band and score it on the upper half it has never seen.

OUT-OF-SAMPLE RMS	THIS FILE	A LINE WITH REAL DISPERSION
constant n_m	0.01227	0.03651
saturating n_m	0.01026	0.00323
cubic β	0.03086	0.01326

On this file n_m moves only $+0.0057$ across the band and the two score alike — any dispersion here is comparable to the noise. On a line that genuinely disperses, the saturating model predicts the unseen half **11× better**. So constant is not wrong here; it is untested here, and it would fail elsewhere. We keep both and report the gap between them as uncertainty.

What his code gets right



The Z_c extraction is exact

$z_0 \cdot \sqrt{((1+S_{11})^2 - S_{21}^2) / ((1-S_{11})^2 - S_{21}^2)}$ is algebraically identical to the ABCD result $\sqrt{B/C}$. Verified numerically across the band: maximum difference $1.4 \times 10^{-2} \Omega$, i.e. floating-point noise. What is lost is only the frequency dependence and the imaginary part, by averaging it to a real scalar — and on data this noisy that averaging is arguably *more* robust than a fitted $Z_c(f)$.



The linear β constraint is the best decision in the file

It cannot produce runaway dispersion. Our own earlier cubic- β fit could, and did: it reached $n_m = 3.24$ at 300 GHz where the data says 2.29, inventing a velocity walk-off of 0.97 against a real 0.02. **That was our bug, and his constraint is what protected him from it.** On a synthetic line with known truth, his n_m at 300 GHz is off by -0.002 ; ours by $+0.023$; the cubic by -0.90 .



The α fit form, and normalising at DC

$a\sqrt{f} + b \cdot f + c$ is the correct skin + dielectric + offset decomposition — we use the same one. And DC is the textbook convention for the EO reference; it is only unsafe because *this particular line rings*.

Three latent bugs that did not bite — yet

In `x_vs_yo`, tested directly:

INPUT CURVE	RETURNS	SHOULD BE
monotonic, crosses -3 dB at 150	150.00	150 ✓
same curve, asked for -4 dB	150.00	200
never reaches -3 dB	300.00	no crossing
ripples across -3 dB	300.00	~ 0

The `yo` argument is accepted and then ignored — `-3` is hardcoded in the body. A curve that never crosses silently returns the end of the sweep as though it were a result. And a curve that ripples across the threshold returns the sweep ceiling. That last one matters: **the moment you mis-terminate the line deliberately — which is exactly what a R_T sweep does — the response ripples and the function silently reports the ceiling.**

Where it leaves us

100.5 GHz, with a defensible band of 99–104 GHz across every fitting choice that survives scrutiny. Neither number is *measured*. The data stops at 80 GHz and the -3 dB point is at $1.3\times$ that, so both codes are reporting the shape of an extrapolation. And the 2.5 mm test pattern shows only 0.88 dB of total insertion loss, which makes α poorly determined no matter who extracts it: over the top quarter of the band it scatters by 32% of its own mean.

The honest conclusion is not "his number is wrong by 17 GHz". It is that **three independent choices in his chain each push the same direction**, two of them with a demonstrable mechanism, and that a test pattern this short cannot support three significant figures from anyone. If the number needs to be defended to a customer, the fix is a longer test structure or a multiline/TRL set — not a better fit.

Every figure here was produced by running both codes on DESIGN_JEREZ_V1_suspe_siliconetch_275.s2p, and by auditing both extractions against a synthetic line whose α , n_m and Z_c are known exactly. The traveling-wave transfer function is byte-identical between the two implementations: maximum difference 8.1×10^{-15} dB.